Shot Quality in Mixed Doubles Curling

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ABSTRACT

Curling has continued to grow as a sport, resulting in new categories entering competition on the Olympic stage. At the forefront, Mixed Doubles Curling has emerged as a new and exciting format that blends elements of the traditional format with changes that encourage a fast paced, high scoring game that proves to be an excellent spectator sport.

With this focus on the sport, curling analytics have been able to grow alongside it and following a landmark collection of data in recent years, there are more frontiers than ever to explore within the field (Ritchie, Leblanc, and Loughin, 2025). Utilizing this dataset, we investigate the application of the shot quality metric in a quantitative format as opposed to the current qualitative metric. After deep investigation and manipulation of the dataset, we applied a variety of machine learning techniques to examine the quantitative changes between different shot tasks and what makes each shot successful. These findings then culminated in a final model that adequately predicted shot quality at 90% accuracy.

This investigation showcased the importance of the status of the playing field and how specific stones drive task selection and success in the match. Distances and stone travel are a critical component to deciding shot task. Our findings indicate that in order to improve analysis, state-space models and sequence-based models should be utilized to build context for predictive efforts.

INTRODUCTION

The recent surge of sport analytics has fueled the utilization of data science models and algorithms within a wide variety of sports. These models allow teams, players, media, viewers, and analysts to evaluate performance in a deeper way than is possible by simple observation of game play or consolidation of box-score statistics. In curling, this shift has been particularly notable with the emergence of Mixed Doubles as a competitive format. Due to the faster pace of this format, decision making processes often happen quicker and involve higher stakes than in traditional curling. These aspects of the game make mixed doubles especially well-suited for quantitative analysis.

A central challenge in curling analytics is the evaluation of shot quality. Because “quality” is a subjective measure, it can be to difficult to grade without human bias being introduced. Scoring outcomes can be observed directly but judging whether a shot successfully achieved its intended objective is more ambiguous. Current approaches rely on human judges, which, as mentioned, create variability and inconsistencies across shots or games. This makes it difficult to systematically compare performance or identify the factors that contribute to successful execution.

The availability of detailed shot-level data provides an opportunity to analyze this problem and investigate possible solutions. By inspecting shifts in game states between each individual stone, we are able to use these factors to attempt to predict score through machine learning. By applying these models, we hope to test the capabilities of machine learning models by searching for patterns within the data that may be indicative of shot success. This is an inherently difficult task but remains consistent and unbiased.

This study ultimately explores how shot characteristics and game context can be used to evaluate shot quality and aid decision making in mixed doubles curling.

RELATED WORK

We based our work on previous research within the sport analytics industry with the biggest influence being Ritchie, Leblanc, and Loughin (2025). Their work provided the dataset used for this analysis along with preliminary investigations of a wide range of curling metrics. The work provided significant insight across all 4 levels of granularity covered in the dataset. Due to this broad approach, however, we are able to identify areas for further investigation, especially in regard to the shot-by-shot analysis.

Currently, there is further research being done by Suzumura et. Al (2026) to extract stone positions from images with an accuracy of 99.37% across over 100,000 images. This development of methodology will lead to further focus on tracking data within curling analytics.

In the broader sports world, Hoffmeister & Rambeau investigated state-space models and Markov decision processes to divide games into discrete "states" to evaluate best decisions. Although curling has defined states within ends, the overall sequencing of shots demands further contextualization for future investigations.

In a different style, Brill, Yurko, and Wyner chose to analyze win-probability models to improve overall decision making. Synthesizing the appropriate game context to gather an accurate win probability is a difficult task but one that can be overcome through further data collection and management along with reasonable modeling efforts.

METHODS

Data

This dataset was gathered from Mixed Doubles Curling events from Ritchie, Leblanc, and Loughlin covering events from 2016 to 2025. This effort gathered numerous different datasets but the most important for this project were the Ends and Stones datasets.

The Ends dataset provides the results for each end of the matches. This includes the points scored for the respective team in each end along with the presence of a power play for that end. A power play creates defined positions for the stones regardless of previous actions, meaning that for every power play, the throwers are seeing the same stone layout.

The Stones dataset provided stone by stone data for the events covered which provided around 26,000 observations. This includes X and Y coordinate pairs that designate the placement of the stone on the ice in relation to the boundaries of the playing field. These pairs were tied to stones numbered 1 through 12, corresponding to the 6 stones thrown by each team in their assigned order. Alongside the coordinate pair information, this dataset included the variables: Task, Handle, and Points, all of which were relevant to this analysis. Task designates the intended objective of the shot.

Draw: aims to stop inside or in front of the "house" (target circles) without hitting another stone	**Wick/Soft Peeling**: a glancing blow where the team's stone hits the side of another stone. "Soft peeling" is a gentler version of a wick.
Front: delivered so that it stops in front of the house to act as a barrier or a starting point for a "guard"	**Promotion Take-out:** Team hits one of their stones, which then knocks an opponent's stone out of play.
Guard: its goal is to protect a scoring stone or a specific area of the house from the opponent's shots.	**Take-out:** a high-speed shot intended to strike an opponent's stone and remove it from play entirely.
Freeze: Team stone comes to rest touching an opponent's stone. The opponent cannot "take out" team's stone without also hitting their own.	**Double Take-out:** a single shot that hits and removes two of the opponent's stones from the house.
Raise / Tap-back: hitting a stationary stone to move it toward a better scoring position.	**Clearing:** a high-weight shot meant to remove guards from the path to the house to clear the play area.
Hit and Roll: Team knocks an opponent's stone out of play; their stone "rolls" to a specific, protected spot.	**Through:** passes through the house entirely without touching anything and goes out of play at the back.

Table 1: Curling Tasks and Definitions

Table 1 lists out the different tasks defined in the dataset along with their objectives. Handle designates the curvature of the throw, i.e. whether it was thrown straight, thrown counterclockwise, or thrown clockwise. The curvature of the shot helps determine where it will land on the field and what its intended target may be.

Points is our main variable of interest. In this case, points is not the resulting score of the shot but rather a subjective rating of how well the shot was able to complete its intended objective. These throws are rated on a scale from 0-4 with 4 indicating the throw completed its task successfully and 0 indicating the throw failed its task entirely. This rating is given out by human judges and recorded in scoresheets. This project aims to be able to accurately designate a points rating to a throw through a purely quantitative measure.

Preprocessing

The dataset was fairly clean, however there were several miscoded and uninterpretable records within the data. We elected to remove the miscoded values as they represented only 1% of the total observations. After further investigation, the team decided that the X and Y coordinate pairs would be difficult to accurately engineer and maintain interpretability.

As a result, distance metrics were created for stones that calculate movement from shot to shot. Since each stone is numbered in a way that corresponds to when it is thrown, this allows us to infer further about how the positions are changing actively from throw to throw. This was also done with respect to how power plays and taking out stones change the overall landscape of the shot. Alongside this reformatting of the data, we created another metric to keep track of the number of stones actively in play. The stones actively in play were critical to understanding the objectives each throw was meant to pursue. A high number of stones resulted in massive changes during each throw. If a poor throw was conducted, this could change the landscape of the game drastically.

Additionally, the Power Play variable first designated between the two types of Power Plays as integers, those where stones are placed on the right side. This was transformed into a categorical variable, designating Left, Right, and No power play utilized in the end.

However, because of the nature of the dataset, the high level of talent required to reach this arena of competition results in a very imbalanced dataset. Over 50% of observations in the dataset were rated as a 4 while throws rated a 1 made up only 3%. As a result, we addressed this imbalance in our dataset using oversampling. Random oversampling allowed us to gather a larger dataset of different shot ratings, totaling over 70,000 observations of equal distribution. A more diverse dataset resulted in an increase in performance compared to randomly selected predictions.

Models Used

In order to develop a better understanding of the data, a number of models were applied to the data to achieve three distinct goals: data visualization, baseline modeling, and sophisticated modeling. The models applied to data include t-SNE, Logistic Regression, K-Means Clustering alongside Principal Component Analysis, Random Forest Classification, MLP Neural Network, and LLM-based analysis.

T-SNE Visualization

After oversampling and scaling the data, the t-SNE visualization in Figure 1 shows clear clustering within the data. This indicates that while there are underlying patterns exhibited, they are complex and nonlinear. In addition to the overall clustering, we also implemented a color-coded key of how many points each shot was awarded by the judges (how successfully their task was carried out, separate from game score in this context). There is a clear point spread in many of the clusters, showing the progression of game states has a significant effect on successful versus unsuccessful shots. There are points of concern since there are not clear boundaries between any shot qualities. This is to be expected considering that each task has different criteria for what is considered success. While certain groupings certainly pop out, given the quite large cluster of overlapping point values in the center right, it is likely that this data is missing information to enhance separation of shots.

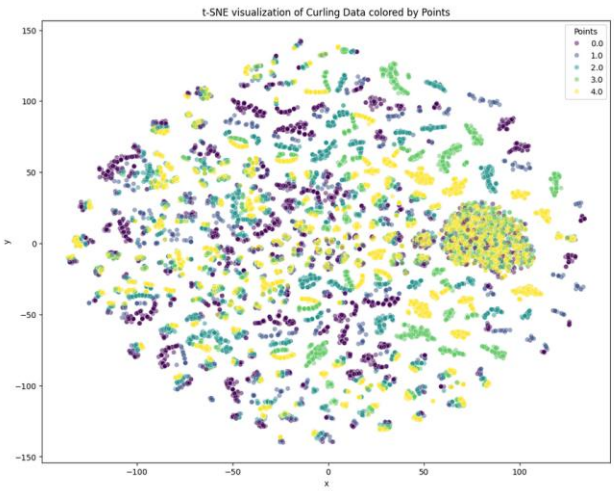


Figure 1: t-SNE Visualization of Curling Shots Colored by Shot Success

Logistic Regression

For our baseline model, we conducted a simple logistic regression. This was our first attempt at a classification, so rather than classifying each shot by individual point values, the binary classification was “successful” (including point values of 3 and 4), and “unsuccessful” (including point values of 0, 1, and 2). There was also an attempt at utilizing a logistic regression model that would predict all 5 values. Although it showed weak performance with the distance features, adding in the stones in play and power play features, we were able to achieve moderately successful results for a baseline in the realm of 0.38. A paired t-test indicated a significant difference compared to random guessing. However, due to the sheer complexity of the data, we knew that improvements could be made.

K-Means Clustering

For further investigation of clustering within shot quality, we applied a K-means clustering model to our stones data. Because our data had very high dimensionality, both the elbow plot and the silhouette score indicated that our optimal number of clusters was twenty. Because this is an unusually high number of clusters for a k-means model, it was very difficult to interpret from a mere glance at a PCA visualization. However, by investigating the properties and contents of each cluster, we were able to gain some meaningful insights. For example, left power play shots were clustered together, right power play shots were clustered together, and a few of the less common tasks (such as double take out) were clustered together.

Random Forest

For our original random forest model, we were able to gain an overarching understanding of which features we needed to pay attention to. However, the initial feature importances also revealed the need for further feature engineering and hyperparameter tuning. To do so, we performed 3-fold cross validation to attain a more sophisticated and robust model. The hyperparameters identified for tuning were the number of trees in model, number of features to consider when looking for the best split, maximum depth of the tree, minimum number of samples to split an internal node, and the minimum number of samples to become at a leaf node. This hyperparameter tuning was able to provide substantial improvements over the initial model.

After analyzing the improved random forest model, we were able to identify the two features with the highest importances: Stones in Play, and Overall Stone Distance. These two features proved to be the strongest predictors of shot quality classification across the dataset.

MLP Neural Network

The MLP neural network model eventually became one of our highest performing models prior to utilizing oversampling. Initially, the model was achieving very high overall accuracies, but this was due to imbalanced data and ghosting of less prevalent point values. This prompted us to seek an optimal balance between metrics including precision, recall, and F1 rather than merely focusing on achieving the highest possible accuracy. To address the imbalance, we scaled and weighted each point value before fitting the model. Additionally, we implemented leaky ReLU activation in an effort to keep all point values present in the results. While this may have lowered our overall accuracy, we were able to achieve a well-fit model (indicated by the accuracy and loss curves shown below) and a more suitable balance between the metrics.

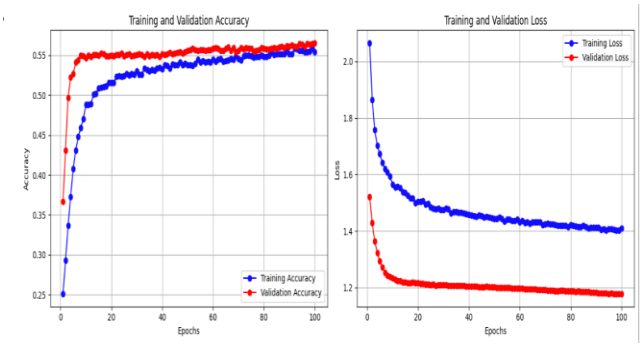


Figure 2: Training/Validation Accuracy and Loss Curves from MLP Neural Network Model

Gemini Assisted Predictions

The final model that we applied to our data was an LLM. We used Google Gemini API for our purposes. Due to the lack of additional context, there was a perceived upper limit to how well the model was able to perform in classification accuracy. However, with adjusted prompting and temperature, the model was able to achieve at a reasonable accuracy when aligned with our specific goals. It is likely that given the relatively new status of Mixed Doubles Curling, the amount of quality information out on the topic is limited and not easily accessible or interpretable with the current Gemini Model. This is especially true at the highest levels of competition where communication of tactics is extremely restricted due to the rapid innovation of different strategies. It is also possible that due to the lack of contextual information included, the model needed to make large, generalized assumptions about the situation.

With a more advanced model, or more information provided, an LLM could possibly prove to be an effective model, but for this scenario, it was not as effective as our random forest and MLP neural network models.

Evaluation Metrics

As previously mentioned, the heavy imbalance in our data prompted us to seek multiple evaluation metrics beyond just overall accuracy. Instead, we sought to achieve a reasonable balance between F1-score and recall of individual classes, as well as total accuracy. However, after completing random oversampling, our ensemble models were able to categorize at a highly effective rate.

Separately, we also used our visualizations to assess the performance of our models. For example, we used the loss and accuracy curves to determine if the model was well-fit, underfit, or overfit, and if the hyperparameters were appropriate for the model. By visually surveying these models, we were able to examine any inaccuracies or behaviors indicative of necessary improvements.

RESULTS

After completing a Post-Hoc Tukey Test, we identified our Random Forest model as the best performing model. The random forest model was able to achieve a test set accuracy of 90.3%. Across other simulations of the model, the model performed slightly worse performing at a rate of 88.4%, significantly better than the Logistic Regression and MLP Neural Network. This was still far beyond the other models, even the MLP neural network which achieved the next highest accuracy of around 55%.

Along with Overall Stone Distance and Stones in Play being identified as critical features, the movement within stones 3 and 9 were high importance features. The state of stone 3 is able to shift the game since it is the first stone thrown after both teams have been able to throw a stone. Stone 9 then signifies another critical stone after 4 ends. Since these stones often have specific tasks of gaining ground from the opposing team, it is clear when there is a failure and when there is success.

DISCUSSION

Future analysis of curling analytics should rely on newly developed methods of gathering high quality positioning data. Following the methodology presented by Suzumura et. Al, there is room for the expansion of curling analytics into the public eye. These ensemble models also provide an interpretable way to dissect decision making in high leverage situations.

There is also further context to be derived from the datasets collected by Ritchie et. Al when considering the teams, players, and tournament context. Given these characteristics, a strength of schedule metric or overall strength rating could be created for the mixed doubles format. To further this research, we would focus more on game states and sequences using Markov chains (Hoffmesiter, Rambau, 2021). This would offer more context within the ends and give insight as to why certain shots are selected and their true goal in order to better evaluate their success.

CONCLUSION

In this project, we explored the capabilities of quantitative analysis to predict shot quality in Mixed Doubles Curling. To address this, we applied a range of machine learning techniques, including k-means clustering, logistic regression, MLP neural network, random forest, and Gemini assisted predictions. This allowed us to classify shot quality with a high accuracy and recall.

Our results indicate that, while there are quantitative patterns in the data, they are complex and difficult to attribute to any few specific predictors. Although total distance and number of stones in play were the most important predictive factors, they do not fully encompass player strategies, skills, or other immeasurable factors. These findings suggest that, while machine learning models can help predict shot quality, the game of curling is more well-suited for complicated state space models.

Future work could expand on this project by focusing on sequential effects and applying a Markov chain model, which is specifically designed for analyzing sequence-dependent state spaces.

Overall, this project demonstrates that although they are imperfect and have limitations, machine learning models are very powerful tools that can provide valuable insights into many different types of data. In Mixed Doubles Curling and the broader world of sports analytics, machine learning is an increasingly utilized tool to understand decision making processes on a deeper level.

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SOFTWARE

Github Repository: https://github.com/davcrown/DSCI303Curling

CONTRIBUTIONS

Rachel implemented k-means clustering and a multilayer perceptron (MLP) neural network as part of the modeling work. In addition, she helped compile the report by writing the introduction, methods, and conclusion sections.

David focused on logistic regression, t-SNE visualization, random forest, and LLM prompting. For the report creation, he was responsible for the abstract, related work, results, and discussion sections.